

		$v \approx 3.3 \text{ ms}^{-1}$
10	B	For mass m_1: $\frac{m_1 v^2}{L + e} = ke$ $v^2 = \frac{ke(L + e)}{m_1}$ For mass m_2: $\frac{m_2 v^2}{2(L + e)} = k(2L + 2e - L)$ $\frac{m_2 v^2}{2(L + e)} = k(L + 2e)$ $m_2 = \frac{2k(L + 2e)(L + e)}{v^2}$ $= \frac{2k(L + 2e)(L + e)m_1}{ke(L + e)}$ $= \frac{2m_1(L + 2e)}{e}$
11	C	total energy at top of ramp = total energy at lowest point